

Question	Marking Guidance	Mark	Comments
6.1	1. (Trend of) slowing growth from before birth to 21 days OR (Trend of) decreasing percentage undergoing mitosis from before birth to 21 days OR (Trend of) decreasing percentage undergoing DNA replication from before birth to 21 days; 2. DNA replication happens before mitosis OR Heart growth slowing until (fully) developed OR These cells lost the ability to divide;	2	1. Accept 'day -6' for 'before birth'. 1. For '21 days' accept 'until the end of the investigation'. 2. Accept 'Heart growing/developing before birth and becomes (fully) developed'. 2. Accept reference to only unipotent cells/cardiomyocytes dividing (at 21 days).
6.2	1. DNA helicase; 2. Breaks hydrogen bonds (between 2 DNA strands); 3. BrdU complementary to <u>adenine</u> (on template strand) OR BrdU forms hydrogen bonds with <u>adenine</u> (on template strand); 4. DNA polymerase joins (adjacent) <u>nucleotides</u> (to incorporate BrdU into the new DNA strand); 5. <u>Phosphodiester bonds</u> form (between nucleotides);	5	2. Reject 'hydrolyses hydrogen bonds' 2 and 3. Accept H bonds for hydrogen bonds. 4. Reject if DNA polymerase catalyses complementary base pairing or if DNA polymerase catalyses nucleotides joining to template strand.

Question	Marking Guidance	Mark	Comments
6.3	1. Add antibody (anti-BrdU with enzyme attached) to cells/DNA OR Add cells/DNA to antibody (anti-BrdU with enzyme attached); 2. Wash (cells/DNA) to remove excess/unattached antibody OR Wash (immobilised antibody) to remove excess/unattached cells/DNA; 3. Add substrate to cause colour change;	3	All mark points must relate to procedure. Do not negate any mark point for use of additional antibodies. 2. Allow ECF for absence of cells/DNA. 3. For 'substrate' accept description in context of enzyme.

